

Di Wang

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Peking University – School of Computer Science – Assistant Professor

Bio

I am an Assistant Professor at Peking University's School of Computer Science, a member of the Programming Languages Lab, and a research advisor of the Turing Program. My main interest is in programming languages, and formal verification, program analysis, and probabilistic programming in particular. My mission is to develop *universal and easy-to-use abstractions and paradigms* for programming safe and efficient software, and *programming-language-level integrations* to automatically analyze, optimize, and synthesize programs. Currently, I am working on resource-safe system programming, programmable Bayesian inference, quantitative program analysis, and proof-oriented programming languages.

Education

Carnegie Mellon University

Ph.D. in Computer Science

Advisor: Prof. Jan Hoffmann

Thesis: *Static Analysis of Probabilistic Programs: An Algebraic Approach*

Pittsburgh, PA, USA

Aug 2017 – May 2022

Peking University

Bachelor of Science (with Honors) in Computer Science & Technology

Advisor: Prof. Yingfei Xiong

Thesis: *Accelerating Program Analyses by Conditional Summarization with Datalog*

Beijing, China

Sep 2013 – Jun 2017

Professional Activities

- **Organizing Committee Member** – APLAS & ATVA (2026 Poster)
- **Program/Review Committee Member** – APLAS (2024 SRC), ASE (2023), ECOOP (2025), ECOOP/ISSTA (2024 Tool Demos), ICALP (2025), ISSTA (2026), OOPSLA (2024, 2026), PLDI (2024), POPL (2026), SLE (2024), TFP (2026), TyDe (2025)
- **Artifact Evaluation Committee Member** – CAV (2020), POPL (2019, 2020)

Awards

- ECOOP Distinguished Reviewer 2025
- OOPSLA Distinguished Reviewer 2024
- SOSPP Best Paper 2025
- OOPSLA Distinguished Paper 2025
- ECOOP Distinguished Paper 2024
- Peking University Boya Fellowship 2024
- China National Scholarship 2014, 2016
- Huawei Scholarship 2015
- Silver Medal (5th place) in the 39th Annual ACM-ICPC World Finals 2015
- Gold Medal (1st place) in the 39th ACM-ICPC Asia Regionals Anshan site 2014
- Gold Medal (9th place) in the 38th ACM-ICPC Asia Regionals Changchun site 2013

Publications

Refereed Conference Papers

- [1] Zhichao Guan, Tailai Yu, **Di Wang**^{*}, and Zhenjiang Hu. Localizing Type Errors for Syntactic Sugar by Lifting. In *Object-Oriented Programming, Systems, Languages, and Applications*, OOPSLA'26, 2026.
- [2] Ziyue Jin and **Di Wang**^{*}. A Program Logic for Under-approximating Worst-case Resource Usage. In *European Symposium on Programming*, ESOP'26, 2026.
- [3] Han Xu and **Di Wang**^{*}. Dependently-Typed AARA: A Non-Affine Approach for Resource Analysis of Higher-Order Programs. In *European Symposium on Programming*, ESOP'26, 2026.
- [4] Junyang Zhang, Xiangcan Xu, Yonghao Zou, Zhe Tang, Xinyi Wan, Kang Hu, Siyuan Wang, Wenbo Xu, **Di Wang**, Hao Chen, Lin Huang, Shoumeng Yan, Yuval Tamir, Yingwei Luo, Xiaolin Wang, Huashan Yu, Zhenlin Wang, Hongliang Tian, , and Diyu Zhou. CortenMM: Efficient Memory Management with Strong Correctness Guarantees. In *Symposium on Operating Systems Principles*, SOSP'25, 2025.
- [5] Qihao Lian and **Di Wang**^{*}. Automatic Linear Resource Bound Analysis for Rust via Prophecy Potentials. In *Object-Oriented Programming, Systems, Languages, and Applications*, OOPSLA'25, 2025.
- [6] Long Pham, **Di Wang**^{*}, Feras A. Saad, and Jan Hoffmann. Programmable MCMC with Soundly Composed Guide Programs. In *Object-Oriented Programming, Systems, Languages, and Applications*, OOPSLA'24, 2024.
- [7] Zhichao Guan, Yiyuan Cao, Tailai Yu, Ziheng Wang, **Di Wang**^{*}, and Zhenjiang Hu. Semantics Lifting for Syntactic Sugar. In *Object-Oriented Programming, Systems, Languages, and Applications*, OOPSLA'24, 2024.
- [8] Ke Sun, **Di Wang**^{*}, Sheng Chen, Meng Wang, and Dan Hao. Formalizing, Mechanizing, and Verifying Class-based Refinement Types. In *European Conference on Object-Oriented Programming*, ECOOP'24, 2024.
- [9] **Di Wang** and Thomas Reps. Newtonian Program Analysis of Probabilistic Programs. In *Object-Oriented Programming, Systems, Languages, and Applications*, OOPSLA'24, 2024.
- [10] Ankush Das, **Di Wang**, and Jan Hoffmann. Probabilistic Resource-Aware Session Types. In *Principles of Programming Languages*, POPL'23, 2023.
- [11] **Di Wang**, Jan Hoffmann, and Thomas Reps. Sound Probabilistic Inference via Guide Types. In *Programming Language Design and Implementation*, PLDI'21, 2021.
- [12] **Di Wang**, Jan Hoffmann, and Thomas Reps. Central Moment Analysis for Cost Accumulators in Probabilistic Programs. In *Programming Language Design and Implementation*, PLDI'21, 2021.
- [13] Tristan Knoth, **Di Wang**, Adam Reynolds, Jan Hoffmann, and Nadia Polikarpova. Liquid Resource Types. In *International Conference on Functional Programming*, ICFP'20, 2020.
- [14] **Di Wang**, David M. Kahn, and Jan Hoffmann. Raising Expectations: Automating Expected Cost Analysis with Types. In *International Conference on Functional Programming*, ICFP'20, 2020.
- [15] Tristan Knoth, **Di Wang**, Nadia Polikarpova, and Jan Hoffmann. Resource-Guided Program Synthesis. In *Programming Language Design and Implementation*, PLDI'19, 2019.
- [16] **Di Wang**, Jan Hoffmann, and Thomas Reps. A Denotational Semantics for Low-Level Probabilistic Programs with Nondeterminism. In *Mathematical Foundations of Programming Semantics*, MFPS'19, 2019.
- [17] **Di Wang** and Jan Hoffmann. Type-Guided Worst-Case Input Generation. In *Principles of Programming Languages*, POPL'19, 2019.

- [18] **Di Wang**, Jan Hoffmann, and Thomas Reps. PMAF: An Algebraic Framework for Static Analysis of Probabilistic Programs. In *Programming Language Design and Implementation*, PLDI'18, 2018.
- [19] Peng Wang, **Di Wang**, and Adam Chlipala. TiML: A Functional Language for Practical Complexity Analysis with Invariants. In *Object-Oriented Programming, Systems, Languages, and Applications*, OOPSLA'17, 2017.
- [20] Hao Tang, **Di Wang**, Yingfei Xiong*, Lingming Zhang, Xiaoyin Wang, and Lu Zhang. Conditional Dyck-CFL Reachability Analysis for Complete and Efficient Library Summarization. In *European Symposium on Programming*, ESOP'17, 2017.

Refereed Journal Papers

- [21] Zhang Cheng, Jiyang Wu, **Di Wang**, and Qinxiang Cao. Denotation-based Compositional Compiler Verification. *Transactions on Programming Languages and Systems*, 48(5), March 2026.
- [22] Ruyi Ji, Yuwei Zhao, Yingfei Xiong*, **Di Wang**, Lu Zhang, and Zhenjiang Hu. Decomposition-Based Synthesis for Applying Divide-and-Conquer-Like Algorithmic Paradigms. *Transactions on Programming Languages and Systems*, 46(8), June 2024.

Refereed Workshop Papers

- [23] Tailai Yu, Zhichao Guan, **Di Wang***, and Zhenjiang Hu. Inferring Typing Rules for Contextual Sugars. In *Partial Evaluation and Program Manipulation*, PEPM'26, 2026.
- [24] **Di Wang**. Composable Effect Handling for Programming LLM-Integrated Scripts. In *Language Models and Programming Languages*, LMPL'25, 2025.

Other Publications

- [25] Changze Huang, **Di Wang**, and Zhi Quan Zhou. LLM-Assisted Model-Based Fuzzing of Protocol Implementations, 2025.
- [26] Zimun Chen and **Di Wang**. PathFuzzing: Worst Case Analysis by Fuzzing Symbolic-Execution Paths, 2025.
- [27] Yiyuan Cao, Jiayi Zhuang, Houjin Chen, Jinkai Fan, Wenbo Xu, Zhiyi Wang, **Di Wang**, Qinxiang Cao, Yingfei Xiong, Haiyan Zhao, and Zhenjiang Hu. C*: Unifying Programming and Verification in C, 2025.
- [28] Changze Huang and **Di Wang**. Incremental Structure Discovery of Classification via Sequential Monte Carlo, 2024.
- [29] Hongjun Wu and **Di Wang**. Worst-Case Analysis is Maximum-A-Posteriori Estimation: Resource Analysis with Sequential-Monte-Carlo-Based Fuzzing, 2023.
- [30] **Di Wang**, Jan Hoffmann, and Thomas Reps. Expected-Cost Analysis for Probabilistic Programs and Semantics-Level Adaption of Optional Stopping Theorems, 2021.

Teaching and Mentoring Experience

- **Lecturer** – *Compiler Principles (Honor Track)*, Peking University S25, S26
- **Lecturer** – *Compiler Principles*, Peking University F23, S24, F24
- **Lecturer** – *Design Principles of Programming Languages*, Peking University S23, S24, S25, S26
- **Lecturer** – *Seminar on Introduction to Computer Systems*, Peking University F23, F24, F25
- **Guest Lecturer** – *Foundations of Quantitative Program Analysis*, Carnegie Mellon University F19
- **PhD Advisor** – Houjin Chen, Jiayi Zhuang 2025–pres.
- **PhD Advisor** – Zimu Chen, Siyuan Zhu 2024–pres.
- **PhD Advisor** – Changze Huang, Qihao Lian 2023–pres.

Talks

Conference Presentations

- Dependently-Typed AARA: A Non-Affine Approach for Resource Analysis of Higher-Order Programs, *ESOP'26*. Apr 2026
- Newtonian Program Analysis of Probabilistic Programs, *OOPSLA'24*. Oct 2024
- Sound Probabilistic Inference via Guide Types, *PLDI'21*. Jun 2021
- Central Moment Analysis for Cost Accumulators in Probabilistic Programs, *PLDI'21*. Jun 2021
- Raising Expectations: Automating Expected Cost Analysis with Types, *ICFP'20*. Aug 2020
- Liquid Resource Types, *ICFP'20*. Aug 2020
- A Denotational Semantics for Low-Level Probabilistic Programs with Nondeterminism, *MFPS'19*. Jun 2019
- Type-Guided Worst-Case Input Generation, *POPL'19*. Jan 2019
- PMAF: An Algebraic Framework for Static Analysis of Probabilistic Programs, *PLDI'18*. Jun 2018

Seminar Presentations

- CStar: Unifying Programming and Verification in C, *Singapore*, Sponsor Invited Talks, *SPLASH 2025*. Oct 2025
- Program Resource Analysis and Verification, *Beijing, China*, *SAVE 2024*. Nov 2024
- Programmable MCMC with Soundly Composed Guide Programs, *Xi'an, China*, *ChinaSoft 2024*. Nov 2024
- A Proof-Oriented Programming Language, *Suzhou, China*, CCF Beautiful Lake Seminar. Aug 2024
- Algebraic Program Analysis of Probabilistic Programs, *Online*. Spring 2024 Seminar Series, *PROBPROG*. Apr 2024
- Algebraic Program Analysis of Probabilistic Programs, *Shanghai, China*, *ChinaSoft 2023*. Dec 2023

Contributed Software

- C*: Unifying Programming and Verification in C <http://cstarlang.org>
- RaML: Resource Aware ML <https://www.raml.co>
- MIRAI: Rust mid-level IR Abstract Interpreter <https://github.com/endorlabs/MIRAI>